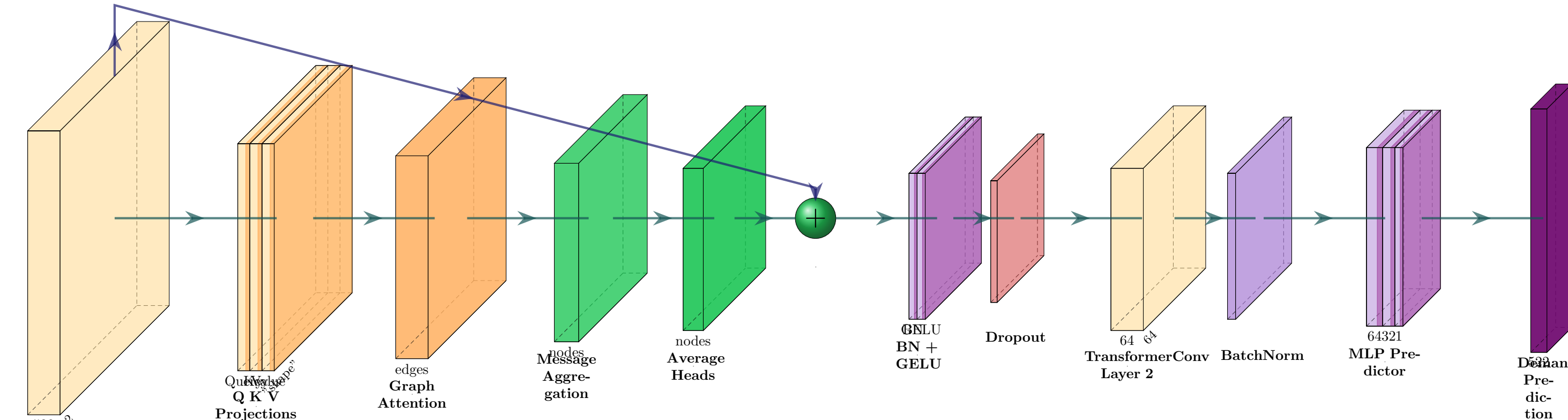


Graph Transformer Architecture v2

Detailed TransformerConv Internal Mechanics



TransformerConv Internal Steps:

1. Node-wise Q,K,V projections: R22 to R64
2. Graph attention: $\alpha = \text{softmax}(Q \cdot K / \sqrt{d})$
3. Message aggregation: $m = \text{sum}(\alpha \cdot V)$
4. Head fusion: Average (concat=False)
5. Residual: $h = m + W \cdot x$
6. Normalization and activation

Tensor Shapes:

Input: x in $\mathbb{R}(522 \times 22)$

Q,K,V: in $\mathbb{R}(522 \times 2 \times 32)$

Attention: α

Messages: m in $\mathbb{R}(522 \times 64)$

Output: $\hat{y}$ in $\mathbb{R}(522 \times 1)$